JULIA CLARICE UY

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LICENSE AND EDUCATION

Geodetic Engineer Licensure Examination, Board Passer September 2026

University of the Philippines Diliman, Quezon City Graduated July 2026
BS Geodetic Engineering, Cum Laude, GWA 1.6883

EXPERIENCE

Geospectrum Marketing Services, Remote Sensing/GIS Intern June 2025 to July 2025

- Mapped sediment plumes and muddy floodwaters in Bulacan and Valenzuela from PlanetScope imagery in QGIS, using the Flood Mud Index on red and blue bands to identify areas needing erosion control near the New Manila International Airport
- Processed HYCOM data to map ocean current magnitude and flow direction in the Philippines, and produced presentation-ready map visuals for public communication
- Performed web GIS development tasks, including map configuration and spatial data integration

GIS AND SPATIAL ANALYSIS PROJECTS

Dasymetric Mapping of Population, Marikina City (ISPRS Annals, PhilGEOS 2025) February 2026

- Classified urban land cover from Sentinel-2 imagery using Random Forest, OBIA, and hybrid OBIA-RF models, achieving 72.19% accuracy for informal settlements, and downscaled census data into a 30 m population grid to reveal high-density clusters not visible on standard census maps
- Prepared the ancillary data, including WorldPop density, PSA census figures, and census blocks, and converted OpenStreetMap river and building vectors into distance rasters

Urban Housing Rent Prediction, Barangay Manggahan, Pasig City (BS Thesis) September 2025 to May 2026

- Developed a Python algorithm that estimates building construction years from Landsat time series, validated on manually traced construction histories; ran Moran's I and local cluster analysis of rent and building age, and produced the study's maps
- Built a Python geocoder using the Google Places API to map web-scraped listings, and trained an XGBoost model with SHAP analysis to identify spatial drivers

Vendor Relocation Spatial Decision Support System, Manila Course project

- Built a web mapping application in React and Leaflet, backed by PostgreSQL/PostGIS, with flood risk and traffic layers and an AHP and Borda voting workflow to aggregate stakeholder priorities

GIS-Based Release Site Analysis for Philippine Eagles, Leyte (Philippine Journal of Science) June 2026

- Processed environmental datasets (DEMs, Human Footprint Index, Key Biodiversity Areas) and applied GIS multi-criteria decision-making with Boolean overlays; wrote and interpreted the findings, which identified about 10,320 ha of suitable release sites

LEADERSHIP

UP Society of Geodetic Engineering Majors (UP GEOP) Member 2023 to 2026

- Academic Deputy, 1st Semester AY 2025 to 2026: coordinated with faculty and the student body on academic affairs and initiatives

SKILLS

- **GIS and Mapping:** QGIS, ArcGIS, ArcMap, Google Earth Engine, OpenStreetMap, geoprocessing, spatial analysis, cartographic map production, remote sensing image processing, SAR and SNAP
- **Spatial Data and Web:** PostgreSQL/PostGIS, geodatabase management, SQL, React, Leaflet, web-based spatial visualization
- **Programming and Other:** Python (XGBoost, SHAP, Moran's I), R, Microsoft Excel, AutoCAD, Civil 3D